

# ECE 447

## Lesson 01

### Course

### Introduction



UNITED STATES  
AIR FORCE  
ACADEMY

# ABOUT ME

# YOUR TURN!

## Introductions

- Go-by name
- Hometown
- AFSC
- Surprise question!

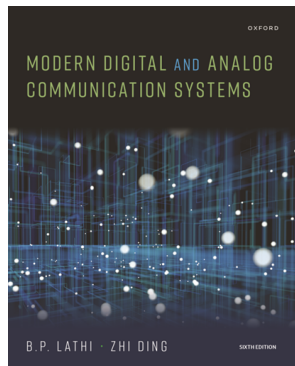
# HOW TO CONTACT YOUR INSTRUCTOR

- Office: 2E36A
- Outlook: Official (e.g., bedrest, SCA, etc.)
- Teams: Questions, problems, concerns
- EI: Prefer Outlook calendar invite (pick any available time) or schedule [here](#) - but walk-ins okay, too! (if available)

# COURSE TEXTBOOK

- *Modern Digital and Analog Communication Systems*, B. P. Lathi and Zhi Ding, Oxford University Press, 2025
- Textbook is required
- Only available digitally
- Discount by using this [link](https://view-su3.highspot.com/viewer/6f8613b4e0d3ad3da4b97879342ea312?iid=6838256713d17dd25a3827f6&source=email.untracked)

(<https://view-su3.highspot.com/viewer/6f8613b4e0d3ad3da4b97879342ea312?iid=6838256713d17dd25a3827f6&source=email.untracked>)



# SYLLABUS AND COURSE CONTENT

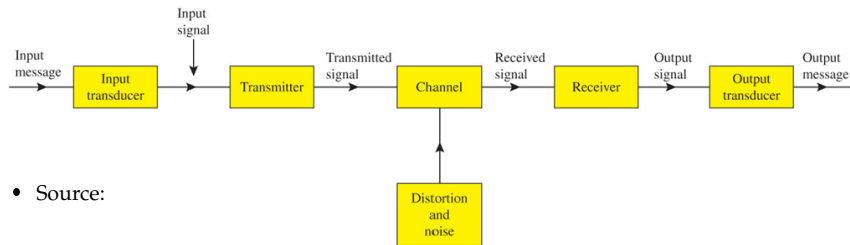
Let's take a look:

<https://usafa-ece.github.io/ece447-book/intro.html>

# WHY THIS COURSE?

- Primary application of EE
- **Focused** on general communication principles
- Many concepts may be familiar from Signal Processing and Linear Systems (ECE 333) and Electromagnetics (ECE 343)
- Will do lots of example problems in class
- Expectation: you **MUST** read the text
- Expectation: you **MUST** do homework problems
- Will have practical applications in computer experiments and Software Defined Radio (SDR) labs
  
- Python or MATLAB?

# COMMUNICATION SYSTEMS

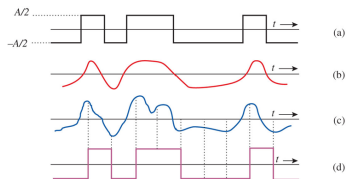


- Source:
- Input transducer:
- Transmitter:
- Channel:
- Noise:
- Receiver:
- Output transducer:
- Destination:



# TERMS

- Channel **bandwidth**: Range of frequencies a channel without too much distortion; should exceed signal BW
- System Analysis
  - **Time and Frequency Analysis**: Use Fourier instead of Laplace, assume steady-state
  - **Deterministic**: Mathematical representation of well-behaved signals
  - **Probabilistic**: Random signals either noise or information signals
- Information Theory and Coding
  - **Shannon Limit**: Information rate vs. channel capacity
  - **Forward Error Correction**: Added redundancy, turbo codes, convolutional codes, block codes



# SHANNON-HARTLEY LAW

Digital Version: The capacity  $C$  of a channel is given by

$$C = BW \cdot \log_2(1 + SNR)$$

where  $C$  is in bits/second or bps,  $BW$  is the bandwidth in Hz, and  $SNR$  is the signal-to-noise ratio in absolute units (not dB).

- Note both sides of equation in units of "per second"
- What parameters can we control?
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